

Top 50 Linux Commands You Must Know

Prepared by: Aditya Ugare

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A comprehensive reference guide to essential Linux commands for system administration, file management, and daily operations.

▮ File and Directory Operations

1. `ls` - List Directory Contents

The `ls` command displays files and directories in the current location.

Syntax: `ls [options] [directory]`

Common options:

- `ls -l` - Long format with detailed information
- `ls -a` - Show all files including hidden files (starting with `.`)
- `ls -lh` - Human-readable file sizes
- `ls -R` - Recursive listing of subdirectories

Example:

```
ls -la /home/user
```

2. `pwd` - Print Working Directory

Displays the absolute path of the current working directory.

Syntax: `pwd [options]`

Example:

```
pwd
```

Output: `/home/user/documents`

3. `cd` - Change Directory

Navigate between directories in the filesystem.

Syntax: `cd [directory]`

Common usage:

- `cd /path/to/directory` - Go to specific directory
- `cd ..` - Move up one directory level
- `cd ~` - Go to home directory

- `cd -` - Return to previous directory

Example:

```
cd /var/log  
cd ~
```

4. `mkdir` - Make Directory

Creates new directories in the filesystem.

Syntax: `mkdir [options] directory_name`

Common options:

- `mkdir -p` - Create parent directories as needed
- `mkdir -m` - Set directory permissions

Example:

```
mkdir -p projects/python/scripts
```

5. `mv` - Move or Rename Files

Moves files/directories to a new location or renames them.

Syntax: `mv [options] source destination`

Common options:

- `mv -i` - Interactive mode, prompt before overwrite
- `mv -v` - Verbose output
- `mv -u` - Move only when source is newer

Example:

```
mv file.txt /backup/  
mv oldname.txt newname.txt
```

6. `cp` - Copy Files and Directories

Copies files or directories from one location to another.

Syntax: `cp [options] source destination`

Common options:

- `cp -r` - Copy directories recursively
- `cp -i` - Interactive mode, prompt before overwrite
- `cp -p` - Preserve file attributes
- `cp -v` - Verbose output

Example:

```
cp -r /source/directory /destination/  
cp file1.txt file2.txt
```

7. rm - Remove Files or Directories

Deletes files and directories from the filesystem.

Syntax: rm [options] file/directory

Common options:

- rm -r - Remove directories recursively
- rm -f - Force removal without prompts
- rm -i - Interactive mode, prompt before each removal
- rm -rf - Force recursive removal (use with caution!)

Example:

```
rm file.txt
```

```
rm -rf old_project/
```

8. touch - Create Empty Files or Update Timestamps

Creates new empty files or updates access/modification times of existing files.

Syntax: touch [options] filename

Example:

```
touch newfile.txt
```

```
touch file1.txt file2.txt file3.txt
```

9. ln - Create Links

Creates hard links or symbolic (soft) links to files.

Syntax: ln [options] source link_name

Common options:

- ln -s - Create symbolic link
- ln -f - Force creation, remove existing destination

Example:

```
ln -s /path/to/original /path/to/link
```

```
ln file.txt hardlink.txt
```

10. clear - Clear Terminal Screen

Clears the terminal screen for better visibility.

Syntax: clear

Example:

```
clear
```

▯ File Viewing and Editing

11. cat - Concatenate and Display Files

Displays file contents, concatenates multiple files, and creates new files.

Syntax: cat [options] filename

Common usage:

- cat file.txt - Display file contents
- cat file1.txt file2.txt - Display multiple files
- cat > newfile.txt - Create new file (Ctrl+D to save)

Example:

```
cat /etc/passwd
```

```
cat file1.txt file2.txt > combined.txt
```

12. echo - Display Text or Variables

Prints text to standard output or writes to files.

Syntax: echo [options] [string]

Common options:

- echo -n - No trailing newline
- echo -e - Enable interpretation of backslash escapes

Example:

```
echo "Hello, World!"
```

```
echo $PATH
```

```
echo "New line" >> file.txt
```

13. less - View File Contents Page by Page

A pager program for viewing file contents with navigation controls.

Syntax: less [options] filename

Navigation:

- Space - Next page
- b - Previous page
- q - Quit
- /pattern - Search forward
- ?pattern - Search backward

Example:

```
less /var/log/syslog
```

14. `man` - Display Manual Pages

Shows detailed documentation for commands and programs.

Syntax: `man [section] command`

Common sections:

- Section 1 - User commands
- Section 5 - File formats
- Section 8 - System administration

Example:

```
man ls  
man 5 passwd
```

15. `uname` - System Information

Displays system information including kernel name, version, and architecture.

Syntax: `uname [options]`

Common options:

- `uname -a` - All information
- `uname -r` - Kernel release
- `uname -m` - Machine hardware name

Example:

```
uname -a
```

Output: Linux hostname 5.15.0-56-generic
x86_64 GNU/Linux

16. `whoami` - Display Current User

Shows the username of the current user.

Syntax: `whoami`

Example:

```
whoami
```

Output: john

17. tar - Archive Files

Creates, extracts, and manages tar archive files.

Syntax: tar [options] archive_name files

Common options:

- tar -czf - Create compressed gzip archive
- tar -xzf - Extract gzip archive
- tar -tzf - List contents of gzip archive
- tar -cjf - Create bzip2 archive
- tar -xjf - Extract bzip2 archive

Example:

```
tar -czf backup.tar.gz /home/user/documents
tar -xzf archive.tar.gz
```

18. grep - Search Text Patterns

Searches for patterns in files using regular expressions.

Syntax: grep [options] pattern [file]

Common options:

- grep -i - Case-insensitive search
- grep -r - Recursive search in directories
- grep -n - Show line numbers
- grep -v - Invert match (show non-matching lines)
- grep -c - Count matching lines

Example:

```
grep "error" /var/log/syslog
grep -r "TODO" /home/user/projects/
```

19. head - Display Beginning of Files

Shows the first part of files (default: 10 lines).

Syntax: head [options] filename

Common options:

- head -n 20 - Display first 20 lines
- head -c 100 - Display first 100 bytes

Example:

```
head /etc/passwd
head -n 5 file.txt
```

20. tail - Display End of Files

Shows the last part of files (default: 10 lines).

Syntax: tail [options] filename

Common options:

- tail -n 20 - Display last 20 lines
- tail -f - Follow file updates in real-time
- tail -F - Follow file with retry

Example:

```
tail /var/log/syslog
```

```
tail -f /var/log/apache2/access.log
```

21. diff - Compare Files

Compares files line by line and shows differences.

Syntax: diff [options] file1 file2

Common options:

- diff -u - Unified format
- diff -c - Context format
- diff -r - Recursive comparison of directories

Example:

```
diff original.txt modified.txt
```

```
diff -u file1.txt file2.txt
```

22. cmp - Compare Files Byte by Byte

Compares two files byte by byte.

Syntax: cmp [options] file1 file2

Common options:

- cmp -b - Show differing bytes
- cmp -l - Show byte position and values

Example:

```
cmp file1.txt file2.txt
```

23. comm - Compare Sorted Files

Compares two sorted files line by line.

Syntax: comm [options] file1 file2

Output columns:

- Column 1: Lines unique to file1
- Column 2: Lines unique to file2

- Column 3: Lines common to both

Common options:

- comm -1 - Suppress column 1
- comm -2 - Suppress column 2
- comm -3 - Suppress column 3

Example:

```
comm sorted1.txt sorted2.txt
comm -12 file1.txt file2.txt # Show only common lines
```

24. sort - Sort Lines of Text

Sorts lines of text files alphabetically or numerically.

Syntax: sort [options] filename

Common options:

- sort -r - Reverse order
- sort -n - Numeric sort
- sort -u - Remove duplicates
- sort -k - Sort by specific field

Example:

```
sort names.txt
sort -n numbers.txt
sort -t: -k3 -n /etc/passwd
```

25. export - Set Environment Variables

Sets or exports environment variables to child processes.

Syntax: export VARIABLE=value

Example:

```
export PATH=$PATH:/new/path
export JAVA_HOME=/usr/lib/jvm/java-11
```

26. zip - Compress Files

Creates compressed zip archives.

Syntax: zip [options] archive.zip files

Common options:

- zip -r - Recursive compression of directories
- zip -e - Encrypt archive
- zip -9 - Maximum compression

Example:

```
zip archive.zip file1.txt file2.txt
zip -r project.zip project_directory/
```

27. unzip - Extract Compressed Files

Extracts files from zip archives.

Syntax: unzip [options] archive.zip

Common options:

- unzip -l - List archive contents
- unzip -d - Extract to specific directory
- unzip -o - Overwrite files without prompting

Example:

```
unzip archive.zip
```

```
unzip -d /destination/path archive.zip
```

28. ssh - Secure Shell Remote Login

Connects to remote systems securely over the network.

Syntax: ssh [options] user@hostname

Common options:

- ssh -p - Specify port number
- ssh -i - Use specific identity (private key) file
- ssh -X - Enable X11 forwarding

Example:

```
ssh user@192.168.1.100
```

```
ssh -p 2222 admin@example.com
```

```
ssh -i ~/.ssh/id_rsa user@server.com
```

29. service - Manage System Services

Controls system services (start, stop, restart, status).

Syntax: service service_name action

Common actions:

- start - Start the service
- stop - Stop the service
- restart - Restart the service
- status - Check service status

Example:

```
sudo service apache2 start
```

```
sudo service mysql status
```

```
sudo service nginx restart
```

30. ps - Process Status

Displays information about active processes.

Syntax: ps [options]

Common options:

- ps aux - Show all processes with detailed information
- ps -ef - Full format listing
- ps -u user - Show processes for specific user

Example:

```
ps aux
ps aux | grep apache
ps -ef | grep mysql
```

⚙ System Administration

31. kill and killall - Terminate Processes

Sends signals to processes to terminate or control them.

kill Syntax: kill [signal] PID

killall Syntax: killall [options] process_name

Common signals:

- -9 or -KILL - Force kill immediately
- -15 or -TERM - Graceful termination (default)
- -1 or -HUP - Hangup signal (reload configuration)

Example:

```
kill 1234
kill -9 5678
killall firefox
killall -9 chrome
```

32. df - Disk Space Usage

Reports filesystem disk space usage.

Syntax: df [options] [filesystem]

Common options:

- df -h - Human-readable format (GB, MB)
- df -T - Show filesystem type
- df -i - Show inode information

Example:

```
df -h
df -hT /home
```

33. mount - Mount Filesystems

Attaches filesystems to the directory tree.

Syntax: mount [options] device directory

Common options:

- mount -t - Specify filesystem type
- mount -o - Mount options (ro, rw, etc.)
- mount -a - Mount all filesystems in /etc/fstab

Example:

```
sudo mount /dev/sdb1 /mnt/usb
sudo mount -t ntfs /dev/sdc1 /mnt/external
mount # List all mounted filesystems
```

34. chmod - Change File Permissions

Modifies file and directory access permissions.

Syntax: chmod [options] mode file

Numeric mode:

- 4 = read (r)
- 2 = write (w)
- 1 = execute (x)

Symbolic mode:

- u (user), g (group), o (others), a (all)
- - (add), - (remove), = (set exactly)

Example:

```
chmod 755 script.sh
chmod u+x file.txt
chmod -R 644 directory/
```

35. chown - Change File Owner

Changes the owner and group of files and directories.

Syntax: chown [options] user:group file

Common options:

- chown -R - Recursive change

Example:

```
sudo chown john:users file.txt
sudo chown -R www-data:www-data /var/www/
```

36. ifconfig - Network Interface Configuration

Configures and displays network interface parameters.

Syntax: `ifconfig [interface] [options]`

Note: `ifconfig` is deprecated; use `ip` command for modern systems.

Example:

```
ifconfig
ifconfig eth0
sudo ifconfig eth0 192.168.1.100 netmask 255.255.255.0
```

37. traceroute - Trace Network Path

Shows the route packets take to reach a network host.

Syntax: `traceroute [options] hostname`

Common options:

- `traceroute -n` - Show IP addresses only (no DNS)
- `traceroute -m` - Set maximum hop count

Example:

```
traceroute google.com
traceroute -n 8.8.8.8
```

38. wget - Download Files from Web

Downloads files from the internet via HTTP, HTTPS, and FTP.

Syntax: `wget [options] URL`

Common options:

- `wget -c` - Continue interrupted download
- `wget -O` - Save with different filename
- `wget -r` - Recursive download
- `wget -b` - Background download

Example:

```
wget https://example.com/file.zip
wget -c https://example.com/large-file.iso
wget -O custom-name.pdf https://example.com/document.pdf
```

39. ufw - Uncomplicated Firewall

Simple firewall management tool for Ubuntu/Debian systems.

Syntax: `ufw [options] rule`

Common commands:

- `ufw enable` - Enable firewall

- `ufw disable` - Disable firewall
- `ufw status` - Show firewall status
- `ufw allow` - Allow connections
- `ufw deny` - Deny connections

Example:

```
sudo ufw enable
sudo ufw allow 22
sudo ufw allow 80/tcp
sudo ufw deny 23
sudo ufw status verbose
```

40. iptables - IP Packet Filter Administration

Advanced firewall administration tool for Linux systems.

Syntax: `iptables [options] chain rule`

Common options:

- `-A` - Append rule to chain
- `-D` - Delete rule from chain
- `-L` - List rules
- `-F` - Flush all rules

Example:

```
sudo iptables -L
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
sudo iptables -A INPUT -s 192.168.1.0/24 -j ACCEPT
sudo iptables -F
```

▯ Package Management

41. apt, pacman, yum, rpm - Package Managers

Package management tools for different Linux distributions.

apt (Debian/Ubuntu):

```
sudo apt update # Update package lists
sudo apt upgrade # Upgrade installed packages
sudo apt install package # Install new package
sudo apt remove package # Remove package
sudo apt search keyword # Search for packages
```

pacman (Arch Linux):

```
sudo pacman -Syu # Update system
sudo pacman -S package # Install package
sudo pacman -R package # Remove package
sudo pacman -Ss keyword # Search packages
```

yum (RHEL/CentOS):

```
sudo yum update # Update packages
sudo yum install package # Install package
```

sudo yum remove package # Remove package
sudo yum search keyword # Search packages

rpm (Red Hat Package Manager):

rpm -ivh package.rpm # Install package
rpm -Uvh package.rpm # Upgrade package
rpm -qa # List all installed packages
rpm -qi package # Query package information

42. sudo - Execute Command as Superuser

Allows permitted users to execute commands as root or another user.

Syntax: sudo [options] command

Common options:

- sudo -u - Run as different user
- sudo -i - Login shell as root
- sudo -s - Shell as root

Example:

sudo apt update
sudo -u postgres psql
sudo -i

43. cal - Display Calendar

Shows a calendar for the current or specified month/year.

Syntax: cal [options] [month] [year]

Common options:

- cal -3 - Show three months (previous, current, next)
- cal -y - Show entire year

Example:

cal
cal 12 2025
cal -y 2025

44. alias - Create Command Shortcuts

Creates shortcuts for longer commands.

Syntax: alias name='command'

Example:

alias ll='ls -la'
alias update='sudo apt update && sudo apt upgrade'
alias ..='cd ..'

To make aliases permanent, add them to ~/.bashrc or ~/.bash_aliases.

45. dd - Convert and Copy Files

Low-level copying and conversion of raw data.

Syntax: dd if=input of=output [options]

Common options:

- bs= - Block size
- count= - Number of blocks to copy
- status=progress - Show progress

Example:

```
sudo dd if=/dev/sda of=/dev/sdb bs=4M status=progress
dd if=/dev/zero of=testfile bs=1M count=100
sudo dd if=ubuntu.iso of=/dev/sdb bs=4M status=progress
```

Warning: Use with extreme caution as it can overwrite entire drives!

46. whereis - Locate Binary, Source, and Manual Files

Locates binary, source code, and manual page files for a command.

Syntax: whereis [options] command

Common options:

- whereis -b - Search only for binaries
- whereis -m - Search only for manual pages
- whereis -s - Search only for source code

Example:

```
whereis python
whereis -b gcc
whereis -m ls
```

47. whatis - Display One-line Manual Description

Shows brief description from manual page.

Syntax: whatis command

Example:

```
whatis ls
```

Output: ls (1) - list directory contents

```
whatis grep
```

Output: grep (1) - print lines that match patterns

48. top - Display System Processes

Real-time display of system processes and resource usage.

Syntax: top [options]

Interactive commands (while running):

- q - Quit
- k - Kill a process
- M - Sort by memory usage
- P - Sort by CPU usage
- h - Help

Example:

```
top
top -u username # Show processes for specific user
```

49. useradd - Create New User

Adds a new user account to the system.

Syntax: useradd [options] username

Common options:

- -m - Create home directory
- -s - Specify login shell
- -G - Add to supplementary groups
- -d - Specify home directory

Example:

```
sudo useradd -m -s /bin/bash john
sudo useradd -m -G sudo,www-data alice
```

50. passwd - Change User Password

Changes password for user accounts.

Syntax: passwd [options] [username]

Common options:

- passwd - Change your own password
- sudo passwd username - Change another user's password
- passwd -l - Lock password (disable account)
- passwd -u - Unlock password

Example:

```
passwd
```

```
sudo passwd john  
sudo passwd -l testuser
```

Conclusion

These 50 Linux commands form the foundation of effective system administration and daily operations in Linux environments. Mastering these commands will significantly improve your productivity and understanding of Linux systems.

Pro Tip: Practice regularly and always refer to manual pages (man command) for detailed information and advanced usage.
